

Лабораторная работа №6

Статическая маршрутизация VLAN

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Цель работы

Настроить статическую маршрутизацию VLAN в сети
и изучить прохождение ICMP-пакетов между виртуальными локальными сетями.

Построение топологии сети

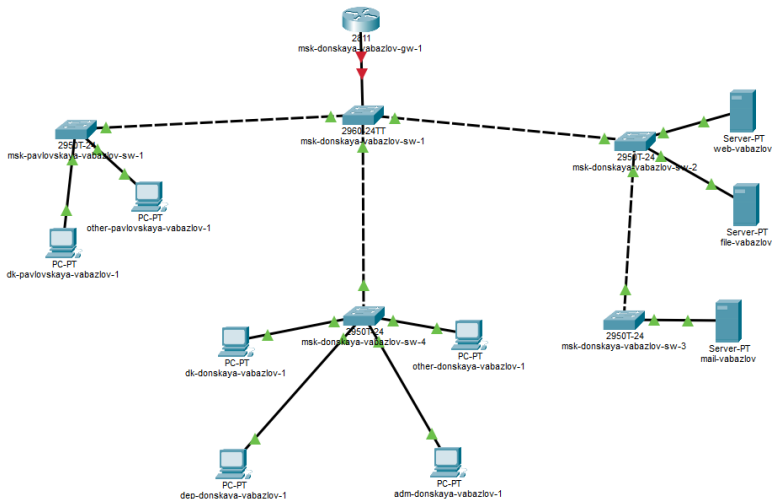
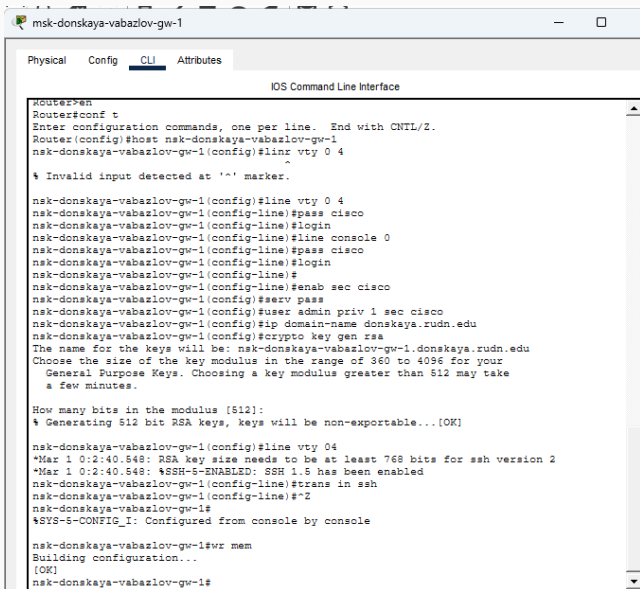


Рис. 1: Топология сети с маршрутизатором и коммутаторами

настройка маршрутизатора



```
msk-donskaya-vabazlov-gw-1
Physical  Config  CLI  Attributes
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host msk-donskaya-vabazlov-gw-1
msk-donskaya-vabazlov-gw-1(config)#linx vty 0 4
^
% Invalid input detected at '^' marker.

msk-donskaya-vabazlov-gw-1(config)#line vty 0 4
msk-donskaya-vabazlov-gw-1(config-line)#pass cisco
msk-donskaya-vabazlov-gw-1(config-line)#login
msk-donskaya-vabazlov-gw-1(config-line)#line console 0
msk-donskaya-vabazlov-gw-1(config-line)#pass cisco
msk-donskaya-vabazlov-gw-1(config-line)#login
msk-donskaya-vabazlov-gw-1(config-line)#
msk-donskaya-vabazlov-gw-1(config-line)#enab sec cisco
msk-donskaya-vabazlov-gw-1(config)#serv pass
msk-donskaya-vabazlov-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-vabazlov-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-vabazlov-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-vabazlov-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-vabazlov-gw-1(config)#line vty 04
*Mar 1 0:2:40.548: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:2:40.548: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-vabazlov-gw-1(config-line)#trans in ssh
msk-donskaya-vabazlov-gw-1(config-line)#*2
msk-donskaya-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-vabazlov-gw-1#
```

Подинтерфейсы Router-on-a-Stick

```
nsk-donskaya-vabazlov-gw-1(config-subif)#int f0/0.102
nsk-donskaya-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

nsk-donskaya-vabazlov-gw-1(config-subif)#enc dot1q 102
nsk-donskaya-vabazlov-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
nsk-donskaya-vabazlov-gw-1(config-subif)#desc departaments
nsk-donskaya-vabazlov-gw-1(config-subif)#
nsk-donskaya-vabazlov-gw-1(config-subif)#int f0/0.103
nsk-donskaya-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.103, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

nsk-donskaya-vabazlov-gw-1(config-subif)#enc dot1q 103
nsk-donskaya-vabazlov-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
nsk-donskaya-vabazlov-gw-1(config-subif)#desc adm
nsk-donskaya-vabazlov-gw-1(config-subif)#
nsk-donskaya-vabazlov-gw-1(config-subif)#int f0/0.104
nsk-donskaya-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.104, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

nsk-donskaya-vabazlov-gw-1(config-subif)#enc dot1q 104
nsk-donskaya-vabazlov-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
nsk-donskaya-vabazlov-gw-1(config-subif)#desc other
nsk-donskaya-vabazlov-gw-1(config-subif)#^Z
nsk-donskaya-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

nsk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
nsk-donskaya-vabazlov-gw-1#
```


Проверка связности между VLAN

Результаты ping

```
dep-donskaya-vabazlov-1
Physical Config Desktop Programming Attributes
Command Prompt

Pinging 10.128.3.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127
Reply from 10.128.3.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.5

Pinging 10.128.5.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127
Reply from 10.128.5.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.6.5

Pinging 10.128.6.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127
Reply from 10.128.6.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Анализ прохождения пакета

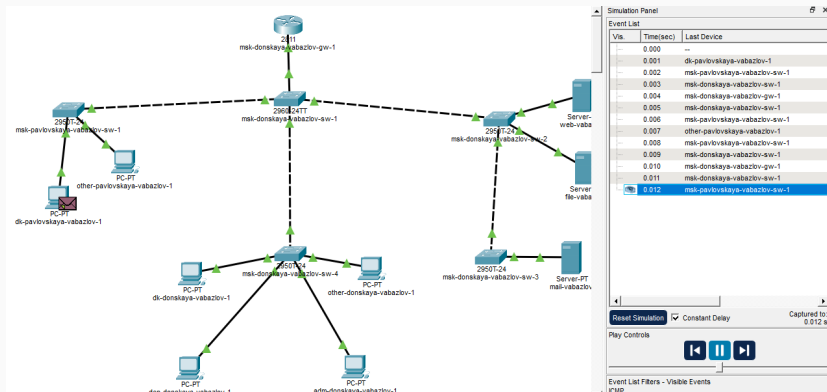


Рис. 5: Передвижение пакета ICMP в режиме симуляции

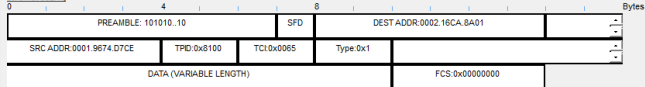
Входящий пакет PDU

PDU Information at Device: msk-donskaya-vabazlov-gw-1

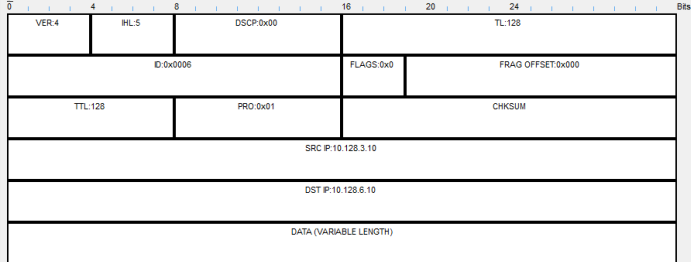
OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

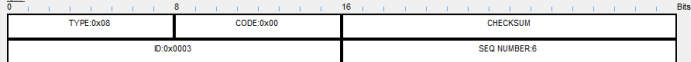
Ethernet 802.1q



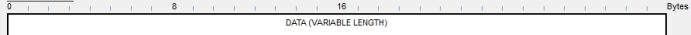
IP



ICMP



Variable Size PDU

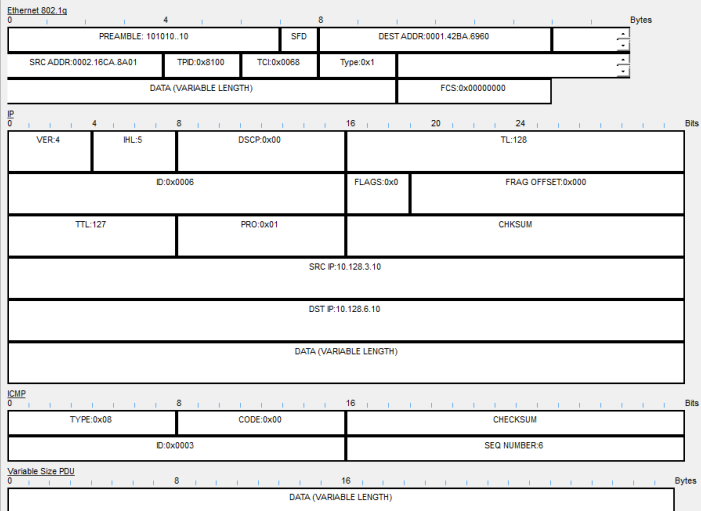


Исходящий пакет PDU

PDU Information at Device: msk-donskaya-vabazlov-gw-1

OST Model Inbound PDU Details Outbound PDU Details

PDU Formats



Итоги работы

В ходе лабораторной работы:

- добавлен и настроен маршрутизатор Cisco 2811;
- настроен доступ к устройству по SSH;
- реализована межвлановая маршрутизация по схеме Router-on-a-Stick;
- назначены IP-адреса виртуальным интерфейсам VLAN;
- проверена связность между узлами разных VLAN;
- изучено прохождение ICMP-пакета в режиме Simulation;
- рассмотрена структура кадров IEEE 802.1Q.